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UNIVERSITY OF PRETORIA
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Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 2: Market Strategic Analysis and Competitors

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

Needs Analysis and its sub-entities were presented. The focus was to adapt the principles of systems design to the design of a business-oriented project by addressing the preliminary conceptual design phase.

Expected Learning Outcomes for the Current Lecture

The expected learning outcomes herein include:

- Gaining understanding, being able to apply, analyse, and evaluate basic principles of strategic marketing such as STP, SWOT analysis, PESTLE analysis and the PORTER's Five Forces in a business-oriented project environment.
- A mastery of the interconnectedness of the dynamic sub-entities that drive the market can be a greater gain towards addressing the market forces.

BLOOM'S TAXONOMY & EXPECTED LEVELS of COGNITION

BLOOM'S TAXONOMY



Presentation Outline

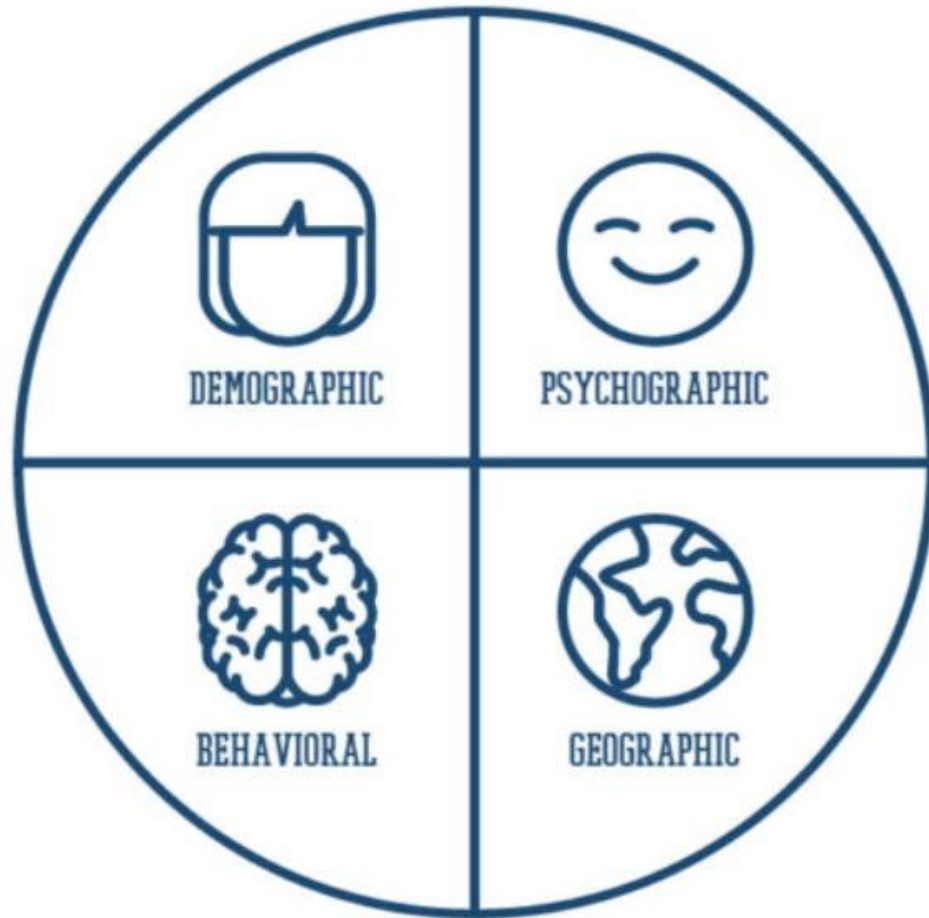
- Market Gap Identification & Exploration
- Business Systems Strategic Analysis
- Market Competitors

Market Gap Identification & Exploration

S – T – P approach

Segmentation → Targeting → Positioning

Segmentation



Segmentation

DEMOGRAPHIC	PSYCHOGRAPHIC	BEHAVIOURAL	GEOGRAPHIC
Age	Lifestyles	Purchasing habits	Urban / rural
Gender	Interests	Spending habits	Radius
Income	Values	User status	Country
Ethnicity	Beliefs	Brand interactions	City/Suburb
Education	Personality traits		Climate

Segmentation

DEMOGRAPHIC	PSYCHOGRAPHIC	BEHAVIOURAL	GEOGRAPHIC
Company size	Values	Purchasing habits	Urban / rural
Industry	Brand	Networking connections	Radius
Job function			Country
			City/Suburb
			Climate

Eight Benefits of Market Segmentation

- Create stronger marketing messages
- Identify the most effective marketing tactics
- Design hyper-targeted ads
- Attract (and convert) quality leads
- Differentiate your brand from competitors
- Build deeper customer affinity
- Identify niche market opportunities
- Stay focused

Targeting

Define a buyer persona

- A buyer persona is a detailed description of someone who represents your target audience. It's often a product of deep research of existing or desired audience.
- It is also referred to as: a customer persona, audience persona, or marketing persona.

Targeting (An example of a customer/ buyer persona)

Description of Ms x:

- she cares so much about her career,
- her appearance, and
- her social life.
- she loves to get ready for her day but doesn't enjoy the amount of time it takes to put on her makeup and fix her hair.

Targeting (An example of a customer/ buyer persona)

Description of Mr y:

- he cares so much about his gym exercise at the expense of lectures.
- he uses a lot of protein shakes for muscle build-up.
- he cannot miss a day without an energy drink.
- he loves to put on Nike sportswear.
- he enjoys travelling around town with uber.
- he eats a lot of pap and soup made from cat fish.

Targeting Define a buyer persona

	They Gain Information By Online articles
Name Sarah	Goals or Objectives • Invest for her future • Save money
Age 25 to 34 years	
Highest Level of Education Associate degree (e.g. AA,)	Motivations • Wants to be financially secure • Wants to retire early
Social Networks     	
Industry Finance	Biggest Challenges • Worrying about the future • Money savings
Organization Size 201-500 employees	

Positioning (Market Mapping)

- Market mapping is a strategy that uses visual aids to organise competitive information, like revenue or growth rate, in order to better understand industry and market trends.
- Market mapping can be used to identify where a business falls within the market, and inform strategic pivots or direction.
- As a small business owner, you understand how crucial it is to know where you stand among your competitors. Learning how to market map and implement it into strategic decision-making can help you find what works well for your business.
- Market mapping refers to the mapping of your competitive landscape.

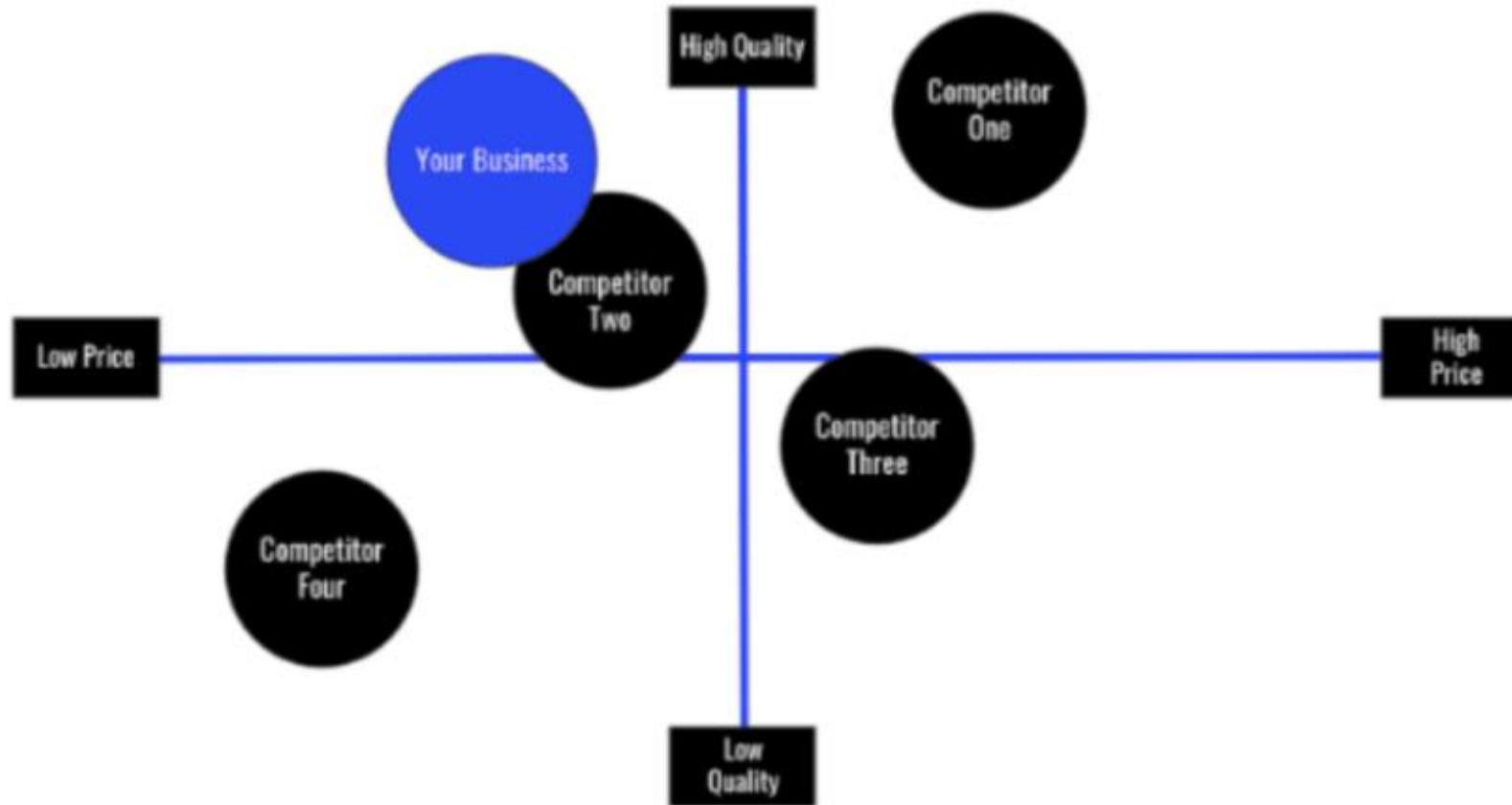
Positioning (Market Mapping Variables)

- High Price vs Low Price
- Low Quality vs High Quality
- Low Tech vs High Tech
- Simple vs Complex
- Short Turnaround vs Long Turnaround
- Low Capability vs High Capability
- Inconvenient vs Convenient
- Unhealthy vs Healthy
- Necessity vs Luxury

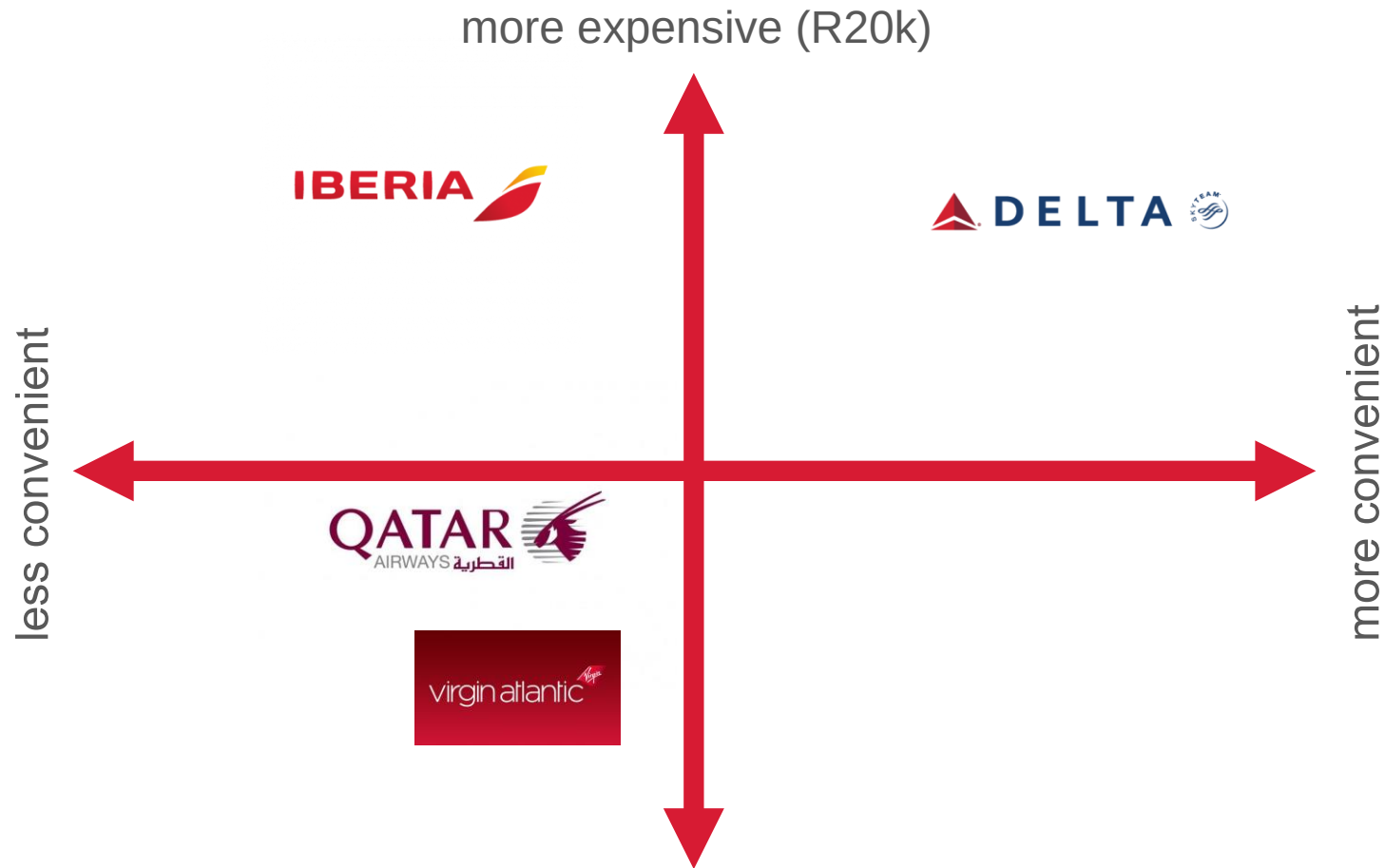
Positioning (Some Marketing Map Categories)

- Maps for Competitors
- Maps for Products
- Maps for Audience

Positioning (Mapping Samples)



Positioning (Mapping Samples)



Positioning (Some Merits of Mapping)

- Creates a visual snapshot
- Displays entire market
- Easy to understand
- Direct comparison with competitors
- Unlimited Variable Combinations
- Provides Quick Market Analysis

Positioning (Some Demerits of Mapping)

- Can be subjective and biased by basing it on sentiments
- Can be deployed without thorough facts, statistics or scientific accuracy yet utilized for the purpose of competition
- Mostly based on two variables (2D driven)

SWOT Analysis for strategic planning of business growth

A SWOT (strengths, weaknesses, opportunities, threats) analysis can help you understand how your business is positioned in relation to the market and your competitors. You can use SWOT analysis for any of the following and many more:

- brainstorming
- strategic business planning
- competitor evaluation
- marketing
- product development
- business reports etc

SWOT Analysis (Internal Factors) Consideration

Strengths	Weaknesses
advantages of the product or business	disadvantages of the product or business
product capabilities	gaps in capabilities
competitive advantages over rivals	competitive weakness versus other similar businesses
innovative aspects of the product/business	outdated aspects of the product/business
any unique selling points or cost of goods	financial constraints or low budget
resources, assets, people	poor processes and systems
experience, knowledge, and data	lack of necessary accreditations
processes, systems, IT and communications	restrictions on IT, systems or communications

SWOT Analysis (External Factors) Consideration

Opportunities	Threats
market developments	political factors
competitors' vulnerabilities	competitors
geographical or export opportunities	environmental pressure
seasonal influences	technological change
current styles or trends	legislative changes
niche markets	customer demand decrease

Advantages of a SWOT analysis

SWOT analysis offers many benefits, including helping you to:

- understand the key issues affecting your business
- spot and deal with weaknesses
- minimise risks and deter threats
- capitalise on opportunities
- take advantage of your strengths
- develop business goals
- identify strategies for achieving your goals

PESTLE Analysis

PESTLE analysis is a fundamental tool for business strategy and planning. It is a method of assessing a business and possible impact on its performance.

PESTLE is an acronym that stands for six external factors that can affect a business viz: political, economic, sociological, technological, legal and environmental. Each of these can have a profound effect on a business in terms of:

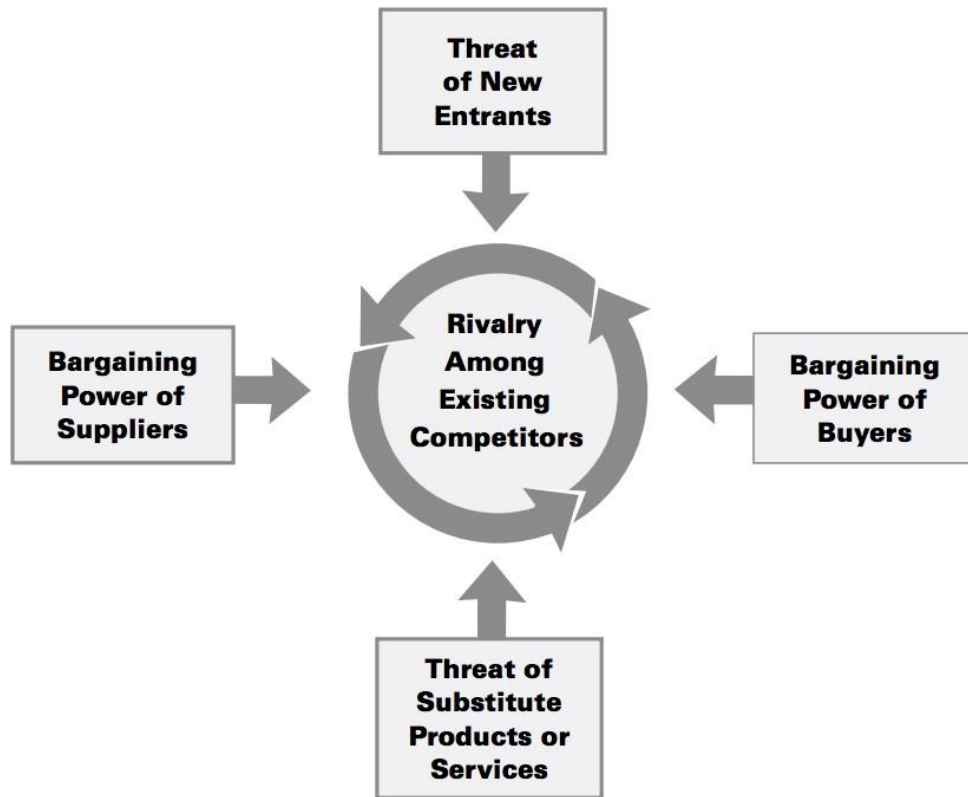
- duration of impact - short term or long term
- type of change - positive, negative or unknown
- rate of impact - increasing, decreasing, unchanged or unknown
- importance - critical, important, unimportant or unknown

Category	Political	Economic	Social	Technological	Legal	Environment
Possible factors	International trading tariffs, restrictions, visa requirements, price control, etc	eg current South African economic situation, currency inflation, interest rates, taxation level, etc	eg cultural norms, attitudes to product, consumer preferences, age and gender distribution, etc	eg emergence of innovative technologies affecting the production, marketing or sale of a product, automation of processes, etc	eg legislative issues, such as consumer protection laws, health and safety laws, licensing regulations, etc	eg sustainability, waste management rules and regulations, green practices etc
Business impact	eg - possible trade barriers to protect domestic suppliers	eg - strength of overseas economies versus South Africa's may affect price/profitability	eg - will product be accepted overseas?	eg - can intellectual property rights be protected overseas?	eg - will the product comply or be allowed into the market?	eg - pollution implications of transportation
Time frame	unknown	6-12 months (possibly longer)	6-12 months	0-6 months	0-6 months	n/a

PESTLE Analysis example for the sale of a product

Category	Political	Economic	Social	Technological	Legal	Environment
Type of impact	negative	unknown	unknown	negative	negative	possibly negative
Rate of impact	increasing	unknown - dependent on the economy, and other countries' economies	unknown	unchanged	unchanged	unknown
Importance	unknown	important	critical	important	critical	unknown

Porter's Five Forces & Business Competitors



Source: Porter, M.E. (2008). The five competitive forces that shape strategy. *Harvard Business Review*. January. Vol 86(1).

How to access:

Google Scholar > search for Five competitive forces that shape strategy > Download pdf from academia.edu

Additional explanation video:

<https://hbr.org/video/3590615226001/theexplainer-porters-five-forces>

Bargaining Power of Suppliers (**Usually High when**):

- **Forward Integration:** The supplier could potentially enter the buyer's industry.
- **Switching Costs:** High costs to switch to another supplier.
- **Concentration:** Few suppliers dominate the industry.
- **Differentiation:** Products are unique or highly differentiated.
- **Substitutes:** No, or few, substitutes exist for the supplier's products.

Bargaining Power of Buyers (**Usually High when**):

- **Backward Integration:** Buyers could potentially produce the product themselves
- **Standardization:** Products are undifferentiated or standardized.
- **Concentration:** Few buyers exist relative to many suppliers.
- **Volume:** Buyers purchase in large volumes, making them important.
- **Switching Costs:** Low costs for buyers to switch to other suppliers.
- **Information:** Buyers are well-informed about market prices and costs.

Threat of Substitute Products/Services: **Major Factors Driving High Threat:**

- **Performance/Price Trade-Off:** The substitute offers better value (lower price or higher performance).
- **Low Switching Costs:** It is easy and cheap for customers to change products.
- **High Similarity:** The alternative offers comparable, or better, utility.

Threat of Substitute Products/Services:

Strategies to Mitigate Threat

- **Differentiation:** Increasing brand loyalty and improving quality.
- **Innovation:** Proactively updating products with new technology.
- **Raising Switching Costs:** Creating ecosystem lock-ins or unique, tailored services.

Threat of New Entrants

- **Impact on Strategy:** Established companies must build and maintain high barriers to entry, such as increasing brand loyalty and investing in proprietary technology, amongst others.
- **High Threat (Low Barriers):** When entering an industry that requires low capital, minimal technology, or has low brand loyalty (e.g., small e-commerce, local cafes), the threat is high.
- **Low Threat (High Barriers):** When there are strict government regulations, high switching costs for customers, high economies of scale, and strong intellectual property, amongst others.

Rivalry Among Existing Competitors: Threat of Old Entrants: **Key Drivers of Intense Rivalry:**

- **Numerous or Equally Balanced Competitors:** Many firms fighting for market share lead to aggressive moves.
- **Lack of Differentiation:** When products are considered commodities, customers switch easily, forcing competitors to compete solely on price.
- **Slow Industry Growth:** As markets mature, growth slows, causing firms to fight fiercely for a larger share of a shrinking or stagnant market.
- **High Fixed Costs:** Firms with high fixed costs are forced to operate at capacity, often slashing prices to maintain sales volume.

Rivalry Among Existing Competitors: Threat of Old Entrants: **Key Drivers of Intense Rivalry:**

- **High Exit Barriers:** Specialized assets or high costs to leave the industry force firms to stay and compete, even if returns are low.

Analysing Porter's Five Forces

- i. What is the hierarchical impact of the five forces on your business?
- ii. What strategies does my business have in place to deal with these forces?
- iii. Synchronisation of SWOT, PESTLE analysis, and PORTER's 5 Forces

Learning Outcomes Evaluated

- Do you have a clear understanding of the concepts of STP, SWOT, PESTLE, & PORTER's 5 Forces?
- Can you apply, analyse and evaluate the outcomes of these strategic market concepts to the semester project for objective market decision making?
- Which of the concepts stood out the most for you?
- You may be familiar with the Gravitational force and other forces in the engineering profession; however, were you amazed that market forces also exist and that they are as real as the forces that dominate the engineering world?

For further reading:

Porter, M.E. (2008). The five competitive forces that shape strategy. *Harvard Business Review*. January. Vol 86(1).

Yesbeck, J. (n.d.) *4 Types of market segmentation with examples.* @Alexa blog. Available at: <https://blog.alexas.com/types-of-market-segmentation/>

<https://www.nibusinessinfo.co.uk>

<https://www.nibusinessinfo.co.uk/content/pestle-analysis-example>

Other Random Sources.

End



Next lecture

Unit 3: Value Chain Design



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